

D849

General Biology I

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

Students model cells as agar cubes in a lab that has new microscopes. One cube is 1 cm on a side and another is 3 cm on a side. The cubes are dyed pink. What is the SA:V ratio of the larger cube?

- A. 6:1
- B. 1:2
- C. 2:1
- D. 18:1

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A cube-shaped cell is 2 μm on a side. What is its SA:V ratio, and is it better or worse at exchange than a 1 μm cube?

- A. 12:1, better than the 1 μm cube.
- B. 3:1, better than the 1 μm cube.
- C. 6:1, the same as the 1 μm cube.
- D. 3:1, worse than the 1 μm cube.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

An enzyme from a heat-loving bacterium found in a famous hot spring speeds up a reaction in the lab. The researchers use a very expensive thermometer. How does the enzyme speed up the reaction?

- A. By raising the temperature of the reaction.
- B. By being used up in the reaction.
- C. By adding energy to the reactants.
- D. By lowering the activation energy.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

During a sprint on a new track, a runner's muscle cells run short of oxygen and produce lactate. The runner wears lightweight shoes. How many ATP per glucose are produced in this pathway compared with aerobic respiration?

- A. About 2, compared with about 30–32 aerobically.
- B. About 30–32, the same as aerobic.
- C. About 36 from fermentation itself.
- D. Zero, because no oxygen is present.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

What is the main role of fermentation in a cell without oxygen?

- A. It runs the electron transport chain.
- B. It produces most of the cell's ATP.
- C. It regenerates NAD⁺ so glycolysis can keep making ATP.
- D. It produces oxygen for the cell.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A student grows a tree seedling in a pot of soil in a sunny classroom window for one year. The tree gains 2 kg, while the soil loses only 50 g. The pot is ceramic. Where did most of the tree's mass come from?

- A. The soil.
- B. Carbon dioxide from the air, fixed in photosynthesis.
- C. Sunlight.
- D. Minerals from the ceramic pot.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

In a cell line studied at a university with a large research budget, DNA is damaged, but the cell keeps dividing because a checkpoint protein is mutated. The lab uses bright fluorescent stains. What is the likely result?

- A. The stains cause the damage.
- B. The cell stops dividing permanently.
- C. The cell becomes healthier.
- D. Uncontrolled division that can lead to a tumor.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A fruit fly has a diploid number of 8 chromosomes. A researcher in a lab with new incubators wants to know how many chromosome combinations are possible in its gametes from independent assortment alone. The flies have red eyes. What is the answer?

- A. 4
- B. 16
- C. 8
- D. 256

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

Independent assortment in a species produces 64 possible chromosome combinations in gametes. What is its diploid number?

- A. 6
- B. 64
- C. 12
- D. 32

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

During childbirth at a hospital with a new maternity wing, contractions trigger the release of oxytocin, which causes stronger contractions and more oxytocin until the baby is born. The hospital has 300 beds. What type of feedback is this?

- A. Positive feedback.
- B. Negative feedback.
- C. Homeostasis through negative feedback, as in all body systems.
- D. A reflex arc.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

When body temperature rises, sweating increases and cools the body back toward normal. What type of feedback is this?

- A. Harmful feedback, because negative means bad.
- B. Negative feedback.
- C. Positive feedback.
- D. No feedback.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A student on a field trip to a scenic desert park writes that cacti grew spines because they needed to protect their water. The park has a popular visitor center. How should the teacher correct this?

- A. Ancestral plants varied; those with spine-like traits survived and reproduced more, so the trait became common over generations.
- B. Each cactus grew spines during its life because it needed them.
- C. Cacti evolved spines because of visitors.
- D. All traits are always perfect adaptations.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

Students model cells as agar cubes in a lab that has new microscopes. One cube is 1 cm on a side and another is 3 cm on a side. ~~The cubes are dyed pink.~~ What is the SA:V ratio of the larger cube?

Target word: compute SA:V **Decides:** $SA = 6 \times 3^2 = 54$; $V = 3^3 = 27$.

A	Wrong stem number	The small cube's ratio.
B	Inverted operation	Volume over surface area.
C	Correct	$54 / 27 = 2$.
D	Dropped element	Uses 6×3 instead of squaring.

2. Answer D Concept 1 • transfer

A cube-shaped cell is $2\ \mu\text{m}$ on a side. What is its SA:V ratio, and is it better or worse at exchange than a $1\ \mu\text{m}$ cube?

Target word: compute and compare **Decides:** SA 24, V 8 $\rightarrow$ 3:1; the $1\ \mu\text{m}$ cube is 6:1.

A	Dropped element	Divides surface area (24) by side length (2) instead of volume (8).
B	Inverted operation	A lower ratio is worse.
C	Wrong stem number	Ratio falls as size increases.
D	Correct	Lower ratio means less efficient exchange.

3. Answer D Concept 2

An enzyme from a heat-loving bacterium ~~found in a famous hot spring~~ speeds up a reaction in the lab. ~~The researchers use a very expensive thermometer.~~ How does the enzyme speed up the reaction?

Target word: enzyme mechanism **Decides:** Enzymes lower activation energy.

A	Camouflage	Mistakes the heat-loving source for the mechanism.
B	Confusion set	Enzymes are reused.
C	Inverted operation	Enzymes do not supply energy.
D	Correct	The core mechanism of catalysis.

4. Answer A Concept 3

During a sprint ~~on a new track~~, a runner's muscle cells run short of oxygen and produce lactate. ~~The runner wears lightweight shoes.~~ How many ATP per glucose are produced in this pathway compared with aerobic respiration?

Target word: ATP yield without oxygen **Decides:** Glycolysis plus fermentation.

A	Correct	Only glycolysis makes ATP; fermentation recycles NAD^+ .
B	Confusion set	That is the aerobic yield.
C	Inverted operation	Fermentation makes no extra ATP.
D	Absolute language	Glycolysis still yields 2 ATP.

5. Answer C Concept 3 • transfer

What is the main role of fermentation in a cell without oxygen?

Target word: purpose, not yield **Decides:** Regenerate NAD^+ so glycolysis can continue.

A	Confusion set	That requires oxygen.
B	Inverted operation	Glycolysis makes the ATP.
C	Correct	The key function of fermentation.
D	Adjacent category	Photosynthesis produces oxygen.

6. Answer B Concept 4

A student grows a tree seedling in a pot of soil in a sunny classroom window for one year. The tree gains 2 kg, while the soil loses only 50 g. ~~The pot is ceramic.~~ Where did most of the tree's mass come from?

Target word: source of mass **Decides:** Soil lost little; mass came from air.

A	Wrong stem number	The soil lost only 50 g.
B	Correct	Carbon from CO_2 builds sugars and tissue.
C	Adjacent category	Light provides energy, not mass.
D	Camouflage	Irrelevant.

7. Answer D Concept 5

In a cell line studied ~~at a university with a large research budget~~, DNA is damaged, but the cell keeps dividing because a checkpoint protein is mutated. ~~The lab uses bright fluorescent stains.~~ What is the likely result?

Target word: result of failed checkpoint **Decides:** Damaged cells divide without control.

A	Camouflage	Irrelevant.
B	Inverted operation	That is what a working checkpoint would do.
C	Confusion set	Cancer cells are not healthier.
D	Correct	Failed checkpoints allow damaged cells to divide.

8. Answer B Concept 6

A fruit fly has a diploid number of 8 chromosomes. A researcher ~~in a lab with new incubators~~ wants to know how many chromosome combinations are possible in its gametes from independent assortment alone. ~~The flies have red eyes.~~ What is the answer?

Target word: combinations **Decides:** $n = 4$, so 2^4 .

A	Dropped element	The haploid number, not combinations.
B	Correct	$2^4 = 16$.
C	Inverted operation	2×4 instead of 2^4 .
D	Wrong stem number	2^8 uses the diploid number.

9. Answer C Concept 6 · transfer

Independent assortment in a species produces 64 possible chromosome combinations in gametes. What is its diploid number?

Target word: solve for chromosome number **Decides:** $2^n = 64$, so $n = 6$ and $2n = 12$.

A	Dropped element	The haploid number.
B	Wrong stem number	The number of combinations.
C	Correct	$n = 6$; diploid = 12.
D	Inverted operation	$64 / 2$.

10. Answer A Concept 7

During childbirth ~~at a hospital with a new maternity wing~~, contractions trigger the release of oxytocin, which causes stronger contractions and more oxytocin until the baby is born. ~~The hospital has 300 beds.~~ What type of feedback is this?

Target word: feedback type **Decides:** The response intensifies the change.

A	Correct	Amplifies until an endpoint.
B	Inverted operation	Would reverse the change.
C	Absolute language	Not all feedback in the body is negative.
D	Adjacent category	A nerve pathway, not hormonal feedback.

11. Answer B Concept 7 · transfer

When body temperature rises, sweating increases and cools the body back toward normal. What type of feedback is this?

Target word: reverse case **Decides:** The response reverses the change.

A	Confusion set	Negative means reversing, not harmful.
B	Correct	It counteracts the rise.
C	Inverted operation	Would make temperature rise further.
D	Dropped element	A sensor-response loop is present.

12. Answer A Concept 8

A student on a field trip ~~to a scenic desert park~~ writes that cacti grew spines because they needed to protect their water. ~~The park has a popular visitor center.~~ How should the teacher correct this?

Target word: correct the need-based view **Decides:** Selection acts on existing variation.

A	Correct	Selection on heritable variation in populations.
B	Inverted operation	Individuals do not evolve in response to need.
C	Camouflage	Irrelevant.
D	Absolute language	Evolution does not produce perfect traits.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	5	
Adjacent category	The right idea in the wrong bucket.	3	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	11	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	4	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Surface-area-to-volume ratio limits cell size	As a cell grows, volume rises faster than surface area, so the SA:V ratio falls and exchange with the environment becomes less efficient.	Thinking bigger cells exchange materials more efficiently because they have more surface.	Cube side lengths and a question about exchange or size.
2	Enzymes lower activation energy	Enzymes speed reactions by lowering activation energy; they are not used up and do not change the overall energy released. Heat or pH extremes can denature them.	Believing enzymes add energy or are consumed.	Reaction rate, temperature, or an energy diagram.
3	Fermentation vs. aerobic respiration	Without oxygen, cells rely on glycolysis plus fermentation, which regenerates NAD ⁺ and yields about 2 ATP per glucose versus about 30–32 with aerobic respiration.	Thinking fermentation itself makes most of the ATP.	Oxygen is absent and ATP yield is asked.
4	Where plant mass comes from	Most of a plant's dry mass comes from carbon dioxide fixed in photosynthesis, not from soil.	Believing plants get their mass from soil.	A plant gains mass while the soil barely changes.
5	Cell cycle checkpoints	Checkpoints halt the cell cycle when DNA is damaged or conditions are wrong; failed checkpoints allow uncontrolled division, as in cancer.	Thinking cancer cells divide because they are healthier.	Damaged DNA and continued division.
6	Independent assortment	The number of chromosome combinations in gametes is 2 ⁿ , where n is the haploid number.	Using the diploid number or multiplying instead of raising to a power.	A chromosome number and a question about combinations.
7	Negative vs. positive feedback	Negative feedback reverses a change to maintain stability; positive feedback amplifies a change until an endpoint is reached.	Thinking negative feedback is harmful or positive feedback is helpful.	A response that reverses vs. intensifies the change.
8	Populations evolve through existing variation	Natural selection acts on existing heritable variation; individuals do not evolve, and traits do not appear because they are needed.	Believing organisms develop traits because they need them.	A claim that an organism changed in response to need.

Course scope (public description): This course is a foundational introduction to the biological sciences. The overarching theories of life from biological research are explored as well as the fundamental concepts and principles of the study of living organisms and their interaction with the environment. Key concepts include how living organisms use and produce energy; how life grows, develops, and reproduces; how life responds to the environment to maintain internal stability; and how life evolves and adapts to the environment.